

Anatole Coulais

Incoming PhD Student in Machine Learning

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📍 Rennes, France

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Education

Exchange	École Centrale de Lyon , Mathematics	Lyon, France
Program	- Stochastic calculus and stochastic differential equations, analysis and modeling of PDEs. - Convex optimization, sparsity, statistics, deep learning, machine learning.	2025 – 2026
Bachelor of Mathematics	Université Claude Bernard Lyon 1 , Mathematics	Lyon, France
	- Probability, measure theory, topology, introductory functional analysis, differential calculus. - Linear algebra, differential equations, group theory, rings and fields.	2022 – 2024
Grandemlyon business school	École , Data Science	Lyon, France
Program	Statistics, artificial intelligence, data science, economics.	2022 – 2026
Classe Préparatoire	Lycée Michel Montaigne , Economics and Mathematics	Bordeaux, France
	Mathematics, economics, philosophy, literature.	2020 – 2022

Experience

CNRS / IRISA , PhD Candidate (incoming)	Rennes, France
PhD supervised by Nicolas Keriven (CNRS) and Titouan Vayer (Inria), on invariance and transferability in generative models for structured data such as graphs and point clouds.	2026 – 2028 2 years
IRISA / Inria, COMPACT team , Research Intern	Rennes, France
Six-month research internship supervised by Nicolas Keriven and Titouan Vayer , on diffusion on graphs and the generalization of generative models for graph machine learning.	Apr 2026 – Sept 2026 7 months
- Invariant stochastic differential equations under group actions. - Equivariant flow matching on sets and graphs, using exchangeability and de Finetti-type structure. - Size generalization for exchangeable generative models via Wasserstein-over-Wasserstein distances.	
Schneider Electric R&D , Data Scientist Intern	Paris, FranceMadrid, Spain
- Co-authored an article on machine learning models of greenhouse gas emissions. - Modeled organic carbon sequestration in agricultural soils. - Co-supervised a student capstone project in data science with Institut Polytechnique de Paris.	Oct 2024 – Aug 2025 1 year
Université de Montréal, Department of Mathematics and Statistics , Research Intern	Montréal, Canada
Supervised by Florian Maire.	Aug 2023 – Dec 2023
- Studied "Unbiased Markov Chain Monte Carlo Methods With Couplings" and implemented the methods in Python. - Presented the paper to about 20 PhD students and researchers in the department.	6 months